$\mathbb{N}, \mathbb{W}, \mathbb{I}, \mathbb{Q}$, Irrational, $\mathbb{R}$.

$\phi = 0$

$\{\phi\} = 1$

$\{\phi, \{\phi\}\} = 2.$

$\{\phi, \{\phi\}, \{\phi, \{\phi\}\}\} = 3$

$S_1 = \{x \in \mathbb{Q} : x^2 < 2\}$

$S_2 = \{x \in \mathbb{Q} : x^2 > 2\}$

there is no max. number in S_1

and there is no min. number S_2.

$x_1 \in S_1$, $x_1^2 < 2$.

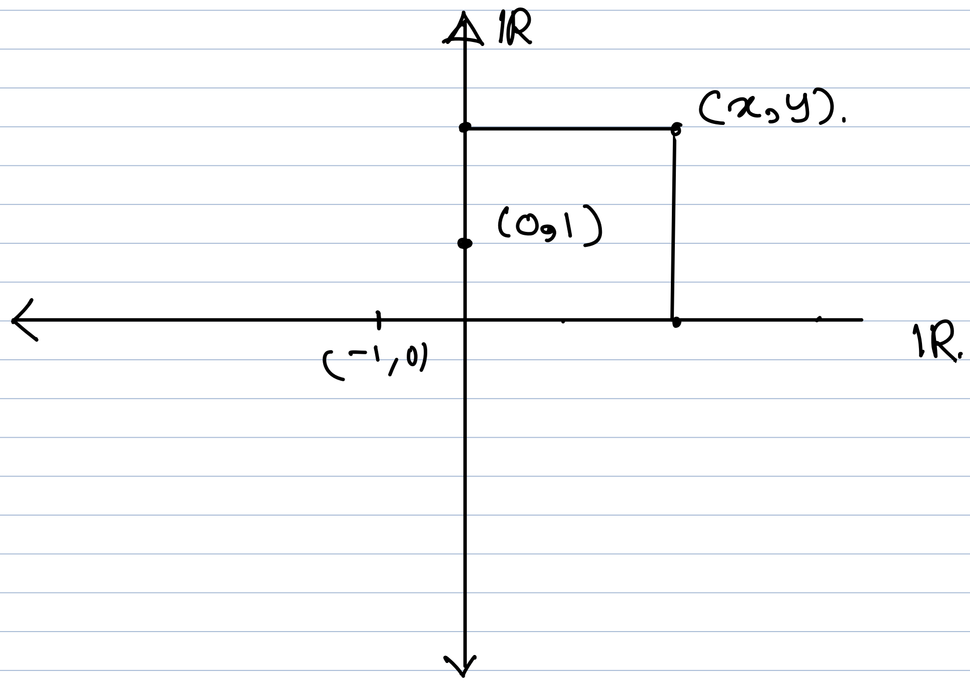
$\forall x \in S_1 \ \exists x' \in S_1 \ \ni x' > x$

$\forall x \in S_2 \ \exists x' \in S_2 \ \ni x' < x$.

$\sqrt{2} = \text{Sup.}(S_1) = \text{inf.}(S_2)$

↳ supremum = least upper bound.

infimum greatest lower bound



Complex Number is an ordered pair of real numbers.

Notation: (x, y) where $x, y \in \mathbb{R}$.

(x, y)

Real Part   Imaginary Part

Let $c_1(x_1, y_1)$ and $c_2(x_2, y_2)$ be two Complex numbers.

$$c_1 + c_2 = (x_1 + x_2, y_1 + y_2)$$

$$c_1 - c_2 = (x_1 - x_2, y_1 - y_2).$$

$$c_1 \times c_2 = (x_1 \cdot x_2 - y_1 \cdot y_2, x_1 \cdot y_2 + x_2 \cdot y_1).$$

$$c_1 / c_2 = \left(\frac{x_1 \cdot x_2 + y_1 \cdot y_2}{x_2^2 + y_2^2}, \frac{y_1 \cdot x_2 - x_1 \cdot y_2}{x_2^2 + y_2^2} \right).$$

$\boxed{i = (0, 1)}$

$i \times i = (\overset{x_1}{0}, \overset{y_1}{1}) \times (\overset{x_2}{0}, \overset{y_2}{1})$

$= (0 \cdot 0 - 1 \cdot 1, 0 \cdot 1 + 0 \cdot 1)$

$= (0 - 1, 0 + 0)$

$= (-1, 0)$

$i^2 = -1$

$\boxed{i = \sqrt{-1}}$

$(\cos\theta + i \sin\theta)^n = \cos(n\theta) + i \sin(n\theta) = e^{in\theta}.$

$\boxed{e^{i\pi} + 1 = 0}$

$C \equiv (x, y)$

$n \in \mathbb{R}.$

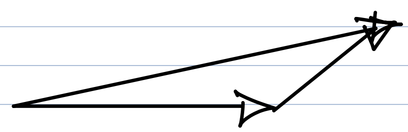
$n \cdot C = (n \cdot x, n \cdot y)$

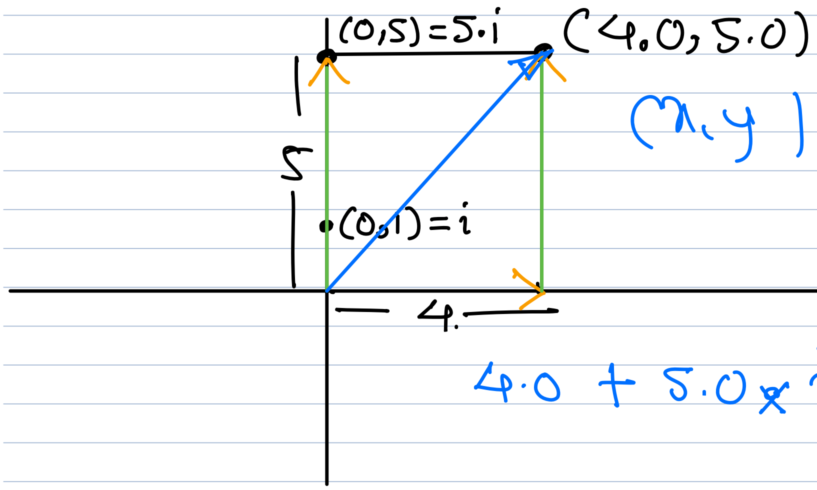
$C = (1.1, 2.2)$

$5 \cdot C = 5 \cdot (1.1, 2.2)$

$= (5, 0) \cdot (1.1, 2.2)$

$= (5 \times 1.1, 5 \times 2.2)$



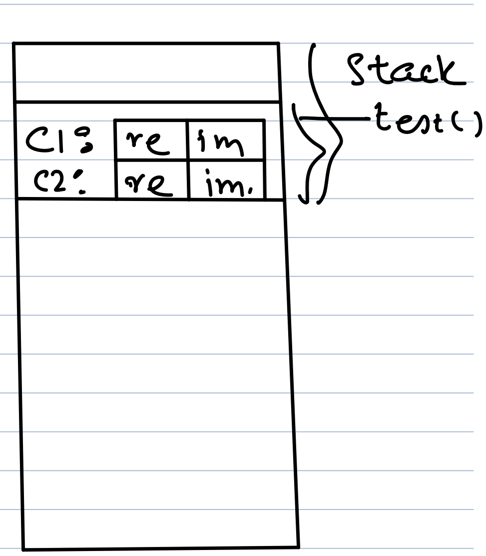


$(x, y) = x + iy$

$4.0 + 5.0 * i =$

```
void test (void)
{
    complex c1 (1.1, 2.2);
    complex c2 (3.3, 4.4);

    // code:
    complex sum = c1.add(c2);
}
```



$x + y \qquad y + x \qquad +, *$

$x * y \qquad y * x$

$x - y \qquad y - x$

$x / y \qquad y / x$

Complex :: add(&c1, c2);

→ complex type of object by value return.

CALLER: c1.add(c2)

Complex::add(&c1, c2)

DEFINITION WRITER:

class complex

{

public:

complex c2

complex& c2

complex add(const complex& c2) const

HIDDEN THIS PTR:

const complex* this

complex add(const complex& c2) const

{

}